

# Self-Distilled Agentic Reinforcement Learning: a Literature-Situated Review

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Study repo: `ai-research-studies/self-distilled-agentic-reinforcement-learning`

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## Notation

A compact symbol table is reproduced from `02-math-deep-dive.md`; full glossary is in that file.

| Symbol                       | Meaning                                                                                      |
|------------------------------|----------------------------------------------------------------------------------------------|
| $\pi_\theta(\cdot \mid s_t)$ | Student policy on student context $s_t$ .                                                    |
| $\pi_T(\cdot \mid s_t^+)$    | Teacher branch: same parameters $\theta$ , privileged context $s_t^+$ .                      |
| $y_t$                        | Token sampled by the student at position $t$ .                                               |
| $\Delta_t$                   | Teacher-student log-prob gap, $\log \pi_T(y_t \mid s_t^+) - \log \pi_\theta(y_t \mid s_t)$ . |
| $h_t$                        | Student entropy, $-\sum_v \pi_\theta(v \mid s_t) \log \pi_\theta(v \mid s_t)$ .              |
| $g_t \in [0, 1]$             | Per-token sigmoid gate (stop-gradient).                                                      |
| $\beta$                      | Gate sharpness (overloaded with the GRPO KL coefficient).                                    |
| $\lambda_{\text{SDAR}}$      | Auxiliary loss weight on the SDAR term.                                                      |

## Background: Multi-Turn Agentic Post-Training

Modern post-training of LLM agents is dominated by reinforcement learning with verifiable rewards. Group Relative Policy Optimization (GRPO), introduced by [Shao et al. \[2024\]](#) and scaled by [DeepSeek-AI \[2025\]](#), is now the workhorse: it strips the value head from PPO [[Schulman et al., 2017](#)] and replaces the GAE baseline with a group-mean baseline computed across  $G$  rollouts of the same prompt. The trajectory-level reward is then broadcast token-wise via a clipped importance ratio. This pipeline is conceptually a descendant of RLHF [[Christiano et al., 2017](#), [Ouyang et al., 2022](#)], with the human-preference reward model swapped for a programmatic verifier.

The structural weakness of GRPO in agentic settings is supervisory sparsity. A 20-turn trajectory in ALFWorld [[Shridhar et al., 2021](#)] or WebShop [[Yao et al., 2022](#)] compresses thousands of token-level decisions into a single scalar reward. *On-Policy Distillation* [[Ye et al., 2026](#), [Yang et al., 2026](#)] and its self-distilled variant *OPSD* [[Zhao et al., 2026](#)] address this by adding a per-token auxiliary loss against a teacher branch. In the self-distilled regime the teacher is the *same* policy conditioned on a privileged context  $s_t^+ = s_t \cup$  (retrieved skill text) available only at training time — a free dense signal. SDAR [[Lu et al., 2026](#)] is the agentic extension of this idea, plus a sigmoid gate to handle the multi-turn failure modes documented below.

## Method: SDAR

The full objective is

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{GRPO}}(\theta) + \lambda_{\text{SDAR}} \cdot \mathcal{L}_{\text{SDAR}}(\theta), \quad (1)$$

with the per-token SDAR loss

$$\ell_t^{\text{SDAR}} = g_t \cdot (\log \pi_T(y_t | s_t^+) - \log \pi_\theta(y_t | s_t)) = g_t \cdot \Delta_t, \quad (2)$$

aggregated by masked-token average,  $\mathcal{L}_{\text{SDAR}} = (\sum_t m_t g_t \Delta_t) / (\sum_t m_t)$ . The gap  $\Delta_t$  is exactly the negative single-sample estimator of the per-token reverse KL between teacher and student, evaluated at the student-sampled token — avoiding a full  $|\mathcal{V}|$ -wide softmax sum.

Lu et al. [2026] benchmark three gating strategies:

$$\text{entropy: } g_t = \sigma(\beta h_t), \quad (3)$$

$$\text{gap: } g_t = \sigma(\beta \Delta_t), \quad (4)$$

$$\text{soft-OR: } g_t = \sigma(\beta [1 - (1 - h_t)(1 - \Delta_t)]). \quad (5)$$

Gap gating is the conceptual centerpiece: it operationalises the asymmetric-trust observation. When  $\Delta_t > 0$  the privileged context is endorsing the student’s sampled token — high-value distillation signal — and the gate opens. When  $\Delta_t < 0$  the disagreement is ambiguous (could be skill-retrieval failure rather than a true student error), and the sigmoid attenuates the update. The gate is detached, so the gradient flows only through the student log-prob:

$$\nabla_\theta \mathcal{L}_{\text{SDAR}} = -\text{Agg}[g_t \cdot \nabla_\theta \log \pi_\theta(y_t | s_t)], \quad (6)$$

which is structurally a gated REINFORCE-style update with  $g_t$  as a per-token signed weight.

## Results

We summarise Table 1 of Lu et al. [2026] on Qwen 2.5 [Qwen Team, 2024].

| Method (Qwen 2.5-3B-Instruct)                                                                                              | ALFWorld $\uparrow$ | Search-QA $\uparrow$ | WebShop $\uparrow$ |
|----------------------------------------------------------------------------------------------------------------------------|---------------------|----------------------|--------------------|
| GRPO baseline                                                                                                              | 75.0                | 36.4                 | 79.8               |
| GRPO + OPSD (ungated)                                                                                                      | —                   | —                    | —                  |
| Skill-GRPO                                                                                                                 | —                   | —                    | —                  |
| Skill-SD                                                                                                                   | —                   | —                    | —                  |
| RLSD                                                                                                                       | —                   | —                    | —                  |
| <b>SDAR (ours)</b>                                                                                                         | <b>84.4</b>         | <b>43.4</b>          | <b>85.0</b>        |
| Numbers reproduced from 02-math-deep-dive.md §8;<br>dashes indicate baseline numbers not transcribed into the study notes. |                     |                      |                    |

The headline gains are +9.4 pp on ALFWorld average success, +7.0 pp on Search-QA [Dunn et al., 2017], and +5.2 pp on WebShop Score for the 3B model. Qwen 2.5-7B-Instruct shows comparable gains. The qualitative companion result is Figure 2 of Lu et al. [2026]: ungated GRPO+OPSD exhibits catastrophic KL-divergence growth across training, while gated SDAR keeps the per-turn KL bounded — direct empirical support for the multi-turn-instability story.

## Position in the Literature

**RL backbone.** SDAR’s GRPO term is taken essentially verbatim from Shao et al. [2024]. The PPO surrogate [Schulman et al., 2017] is therefore the deepest underlying optimisation primitive — with the value head removed and the GAE baseline replaced by a group-mean baseline. SDAR contributes *nothing* to the RL backbone itself; its novelty is the auxiliary objective.

**Distillation lineage.** The auxiliary term descends from On-Policy Distillation [Ye et al., 2026, Yang et al., 2026], which formalised teacher guidance on student-sampled trajectories, and from On-Policy Self-Distillation [Zhao et al., 2026], which collapses teacher and student into a single set of parameters with a privileged-context teacher branch. SDAR’s contribution to this lineage is two-fold: (i) the agentic, multi-turn observation that vanilla OPSD diverges, and (ii) the gating remedy.

**Curriculum vs gating.** A parallel response to multi-turn OPSD instability is the turn-level curriculum of TCOB [Wang et al., 2026], which schedules teacher influence by turn index. SDAR’s gating is the “smooth” alternative — the gate is computed per token from the realised teacher–student gap rather than scheduled exogenously by turn. The two designs are not mutually exclusive; combining them is an obvious follow-up.

**RLHF foundations.** Conceptually the gated distillation term plays a role analogous to the KL-to-reference regulariser in InstructGPT [Ouyang et al., 2022] and the broader RLHF programme [Christiano et al., 2017]: both prevent the policy from drifting too far from a reference distribution. SDAR replaces the static SFT reference with a dynamic privileged-context teacher and gates the constraint per-token rather than treating it as a fixed Lagrangian.

**Sibling agentic-RL work.** Hegde and NovaSky-AI Contributors [2026] (the “Reinforcing Recursive Language Models” alphaXiv blog from the NovaSky-AI / SkyRL group) attacks the same supervisory-sparsity problem from a different angle: a shared policy is rolled out across a *recursive call tree*, and child rollouts inherit advantages from their parent. Credit assignment is by tree topology rather than by per-token reliability gating. The two designs are complementary; § Discussion sketches a hybrid.

**Benchmarks.** The agentic suite is standard: ALFWorld [Shridhar et al., 2021], WebShop [Yao et al., 2022], and Search-QA-style tasks [Dunn et al., 2017]. None of these were designed to stress 50+-turn horizons, which is one reason the depth-scaling question (below) remains open.

## Discussion

**What the experiments support.** Three claims are well-supported by Table 1 and Figure 2 of Lu et al. [2026]: (a) ungated OPSD on top of GRPO destabilises multi-turn training; (b) any of the three proposed gates restores stability; (c) gated SDAR strictly improves over the GRPO baseline on every reported aggregate metric for both 3B and 7B backbones. The KL-bounded behaviour shown in Figure 2 is direct evidence for the failure-mode analysis in § 1 of the paper.

**What is hand-waved.** The bias–variance argument for gating is informal in the paper — there is no formal theorem that gap-gated OPSD is a lower-variance estimator than ungated OPSD, only an empirical demonstration. Sensitivity to the two new hyperparameters ( $\beta$  and  $\lambda_{\text{SDAR}}$ ) is not ablated in the main text. The interpretation of  $\Delta_t$  as a “trust” signal also presupposes that teacher log-prob is a reliable proxy for skill-retrieval quality — a confidently wrong teacher (high  $\Delta_t$ , low actual quality) would be heavily distilled into the student. This is plausible but not directly tested; the slight under-performance of SDAR vs GRPO+OPSD on some Search-QA sub-tasks is consistent with this critique.

**Three future directions.** (1) *Depth scaling*: re-run on a benchmark with 50–100 effective turns and check whether the per-turn KL stays bounded. (2) *Adaptive gates*: learn  $\beta$  jointly with  $\theta$ , or replace the hand-designed gate with  $g_\phi(s_t, \Delta_t, h_t)$ . (3) *Recursive-call extension*: lift gating into the recursive setting of [Hegde and NovaSky-AI Contributors \[2026\]](#) — gate child rollouts by whether the parent endorses them, combining SDAR-style per-token gating with RLM-style tree-level credit assignment.

## See Also

This paper sits inside a broader study chain. The mathematical prerequisites are derived from scratch in [Chapter 25 \(causal LM as MDP\)](#), [Chapter 27 \(tiny GPT pre-training\)](#), [Chapter 28 \(SFT, RLHF, DPO\)](#), [Chapter 30 \(max-entropy RL\)](#), and [Chapter 31 \(PG / GRPO derivation\)](#). The sibling agentic-RL study lives at [pleyva2004/reinforcing-recursive-language-models](#) — it tackles the same supervisory-sparsity problem with a different mechanism (shared-policy + advantage inheritance across a recursive call tree, vs SDAR’s teacher branch + gated per-token distillation), and a hybrid is sketched above.

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